Calculus I Lesson 32

1) Let $f(x) = x^2$.

Find Δy when $x = 2$ and $\Delta x = 0.01$.

a) When $x = 2$, $f(2) =$ ______________

b) $f(x + \Delta x) = f(2 + 0.01) = f(2.01) =$ ____________

c) $\Delta y = f(x + \Delta x) - f(x) = f(2.01) - f(2) =$ ______________

d) $f'(x) =$ __?__ and $f'(2) =$ __?__

e) Let $dx = \Delta x = 0.01$. Find $dy = f'(x)dx = f'(2) \cdot \Delta x =$ ______________

f) How are the values of Δy and dy compared? Is the difference large, small?

2) Let $f(x) = \sin x - 0.5$.

Find Δy when $x = \pi/2$ and $\Delta x = 0.01$.

a) When $x = \pi/2$, $f(\pi/2) =$ ______________

b) $f(x+\Delta x) = f(\pi/2 + 0.01) =$ ____________

c) $\Delta y = f(x+\Delta x) - f(x) = f(\pi/2+0.01) - f(\pi/2) =$ ______________

d) $f'(x) =$ __?__ and $f'(\pi/2) =$ __?__

e) Let $dx = \Delta x = 0.01$. Find $dy = f'(x)dx = f'(\pi/2)\cdot\Delta x =$ ______________

f) How are the values of Δy and dy compared? Is the difference large, small?

3) $f(x) = x^3 + 4$

a) $\dfrac{dy}{dx} = f'(x) =$ ______________

b) $dy = f'(x)dx =$ ______________

Definition of Differentials

a) The differential of x is denoted by dx.

b) The differential of y is denoted by $dy = f'(x) \cdot dx$.

4) $f(x) = x^3 + x - 4$

a) $\dfrac{dy}{dx} = f'(x) =$ ______________

b) $dy = f'(x)dx =$ ______________

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5) $f(x)=\sqrt{x-1}-x+2=(x-1)^{1/2}-4x+2$

a) $\dfrac{dy}{dx}=f'(x)=$ ______________ (multiple choice)

A) $\dfrac{1}{2}(x-1)^{3/2}-4$ B) $\dfrac{1}{4}(x-1)^{2}-4$ C) $\dfrac{1}{2}(x-1)^{-1/2}-4$ D) $(x-1)^{-1/2}-4$

b) $dy=f'(x)dx=$ ________________(multiple choice)

A) $\left[\dfrac{1}{2}(x-1)^{3/2}-4\right]dx$ B) $\left[\dfrac{1}{4}(x-1)^{2}-4\right]dx$ C) $\left[\dfrac{1}{2}(x-1)^{-1/2}-4\right]dx$ D) $\left[(x-1)^{-1/2}-4\right]dx$

6) $f(x) = 7 - x - \sin x$

a) $\dfrac{dy}{dx} = f'(x) =$ ______________

b) $dy = f'(x)dx =$ ______________

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